

MVLU COLLEGE

Subject:-Cyber Security

1.A. Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques

CLI:-

```
substitution_cli.py - C:/Users/mvlu/Desktop/substitution_cli.py (3.14.6)
File Edit Format Run Options Window Help

def encrypt(text, shift):
    result = ""

    for char in text:
        if char.isalpha():
            if char.isupper():
                result += chr((ord(char) - 65 + shift) % 26 + 65)
            else:
                result += chr((ord(char) - 97 + shift) % 26 + 97)
            else:
                result += char

    return result

def decrypt(text, shift):
    return encrypt(text, -shift)

message = input("Enter Message : ")
shift = int(input("Enter Shift Value : "))

encrypted = encrypt(message, shift)
print("Encrypted Message :", encrypted)

decrypted = decrypt(encrypted, shift)
print("Decrypted Message :", decrypted)

Ln: 28 Col: 0
29°C Mostly sunny 09:21 13-07-2026
```

Output:-

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/mvlu/Desktop/substitution_cli.py =====
Enter Message : Vikas Pal
Enter Shift Value : 3
Encrypted Message : Ylndv Sdo
Decrypted Message : Vikas Pal
>>>

Ln: 9 Col: 0
29°C Mostly sunny 09:21 13-07-2026
```

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GUI:-

```
"substitution_cli.py - C:/Users/mvlu/Desktop/substitution_cli.py (3.14.6)"
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import tkinter as tk
from tkinter import messagebox

# Function to encrypt text
def encrypt():
    try:
        text = entry.get()
        shift = int(shift_entry.get())

        result = ""

        for char in text:
            if char.isalpha():
                if char.isupper():
                    result += chr((ord(char) - 65 + shift) % 26 + 65)
                else:
                    result += chr((ord(char) - 97 + shift) % 26 + 97)
            else:
                result += char

        output.config(text="Encrypted: " + result)

        # Replace entry with encrypted text
        entry.delete(0, tk.END)
        entry.insert(0, result)

    except ValueError:
        messagebox.showerror("Error", "Please enter a valid shift value.")

# Function to decrypt text
def decrypt():
    try:
        text = entry.get()
        shift = int(shift_entry.get())

        result = ""

        for char in text:
            if char.isalpha():
                if char.isupper():
                    result += chr((ord(char) - 65 - shift) % 26 + 65)
            else:
                result += char

        output.config(text="Decrypted: " + result)

        # Replace entry with decrypted text
        entry.delete(0, tk.END)
        entry.insert(0, result)

    except ValueError:
        messagebox.showerror("Error", "Please enter a valid shift value.")

root = tk.Tk()
root.title("Caesar Cipher")
root.geometry("400x220")
root.resizable(False, False)

tk.Label(root, text="Enter Message:", font=("Arial", 11)).pack(pady=5)

entry = tk.Entry(root, width=35, font=("Arial", 11))
entry.pack()

tk.Label(root, text="Enter Shift Value:", font=("Arial", 11)).pack(pady=5)
shift_entry = tk.Entry(root, width=10, font=("Arial", 11))
shift_entry.pack()

tk.Button(root, text="Encrypt", width=12, command=encrypt).pack(pady=5)
tk.Button(root, text="Decrypt", width=12, command=decrypt).pack(pady=5)

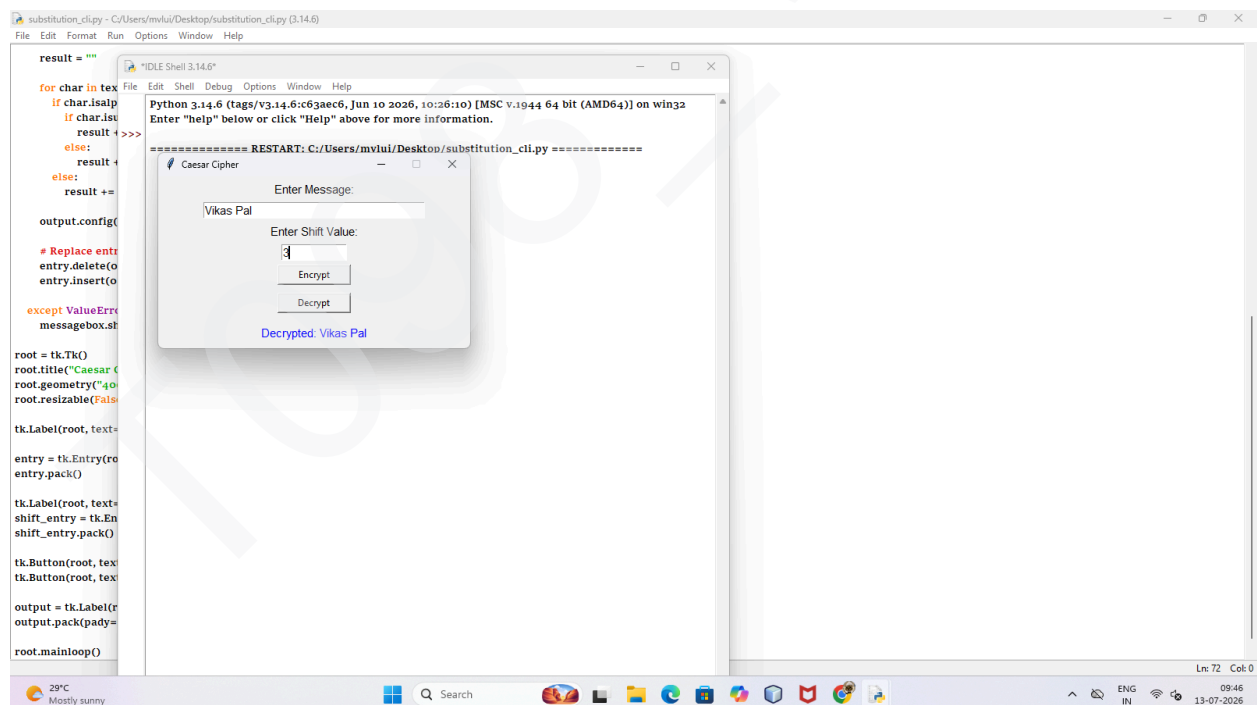
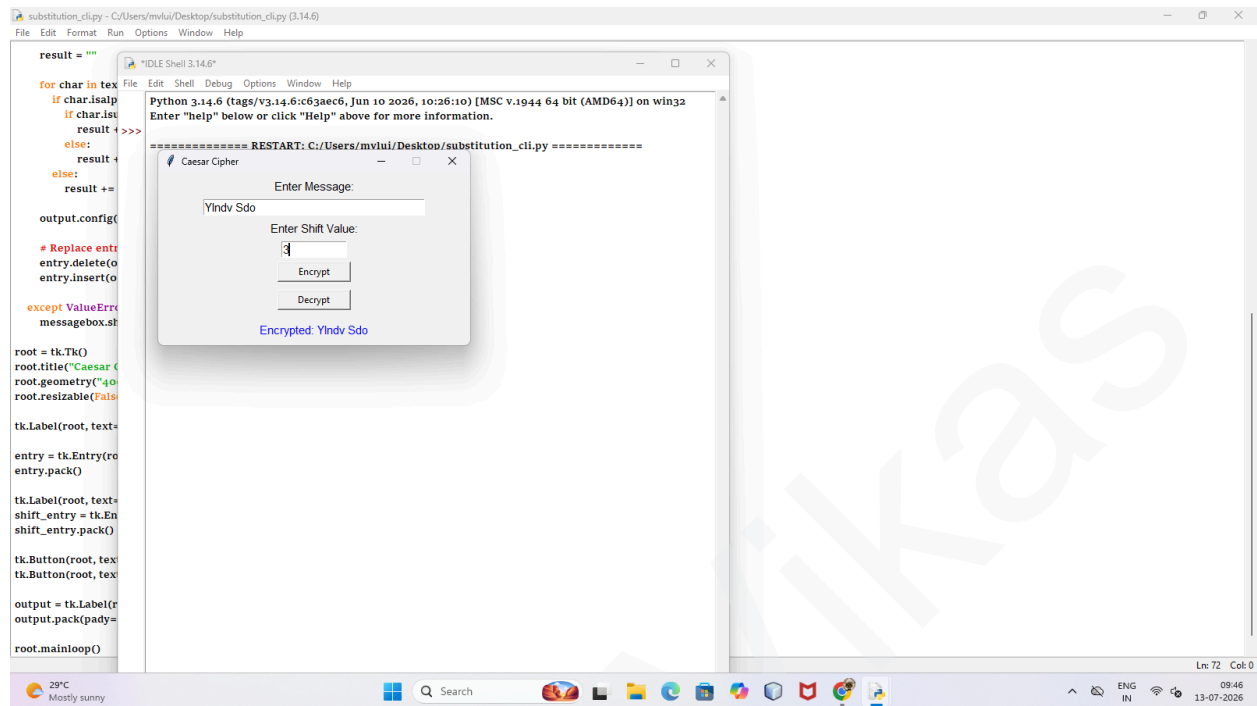
output = tk.Label(root, text="", font=("Arial", 11), fg="blue")
output.pack(pady=10)

root.mainloop()
```

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Output:-



Vikas Pal
T098

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1. B. Design and implement algorithms to encrypt and decrypt messages using transposition techniques

CLI:-

```
*substitution_cli.py - C:/Users/mvlu/Desktop/substitution_cli.py (3.14.6)
File Edit Format Run Options Window Help

def encrypt(text, key):
    rail = [''] * key
    direction = 1
    row = 0

    for ch in text:
        rail[row] += ch

        if row == 0:
            direction = 1
        elif row == key - 1:
            direction = -1

        row += direction

    return ''.join(rail)

def decrypt(cipher, key):
    rail = [['\n' for i in range(len(cipher))] for j in range(key)]

    direction = None
    row = 0
    col = 0

    for i in range(len(cipher)):
        if row == 0:
            direction = 1
        if row == key - 1:
            direction = -1

        rail[row][col] = '*'
        col += 1
        row += direction

    index = 0
    for i in range(key):
        for j in range(len(cipher)):
            if rail[i][j] == '*' and index < len(cipher):
                rail[i][j] = cipher[index]
                index += 1

    result = []
    row = 0
    col = 0

    for i in range(len(cipher)):
        if row == 0:
            direction = 1
        if row == key - 1:
            direction = -1

        result.append(rail[row][col])
        col += 1
        row += direction

    return ''.join(result)

text = input("Enter Message : ")
key = int(input("Enter Key : "))

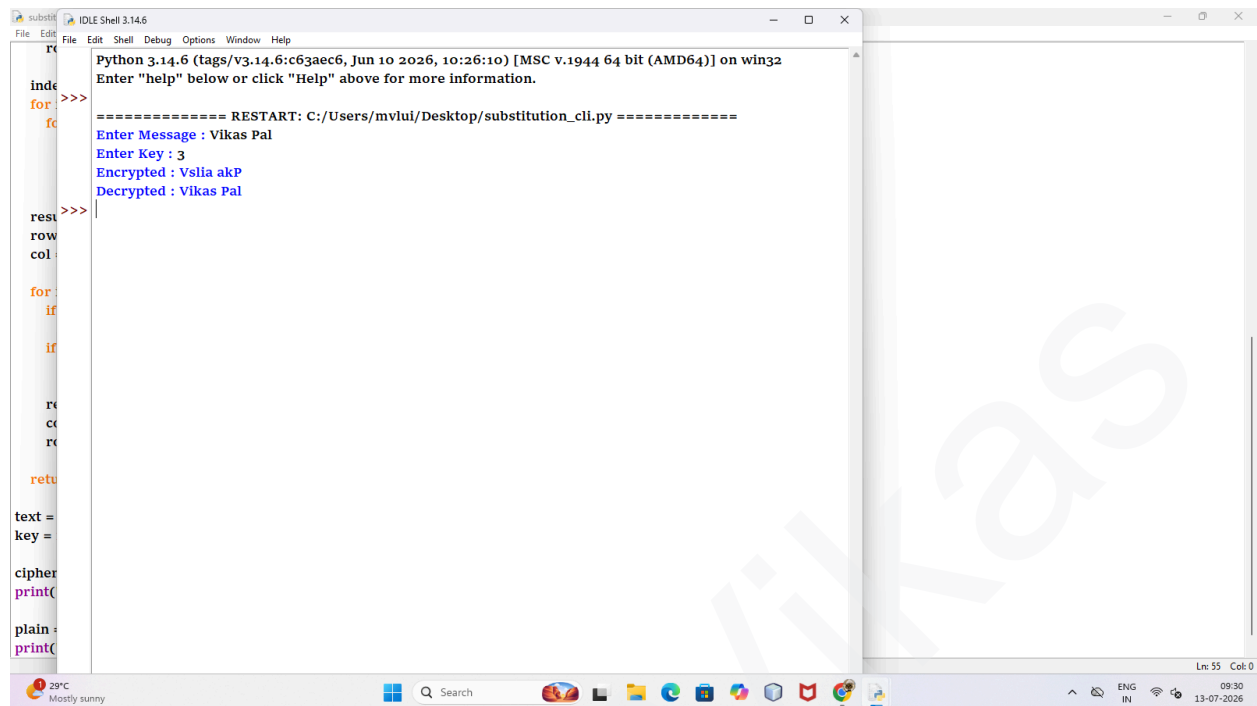
cipher = encrypt(text, key)
print("Encrypted :", cipher)

plain = decrypt(cipher, key)
print("Decrypted :", plain)
```

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Output:-

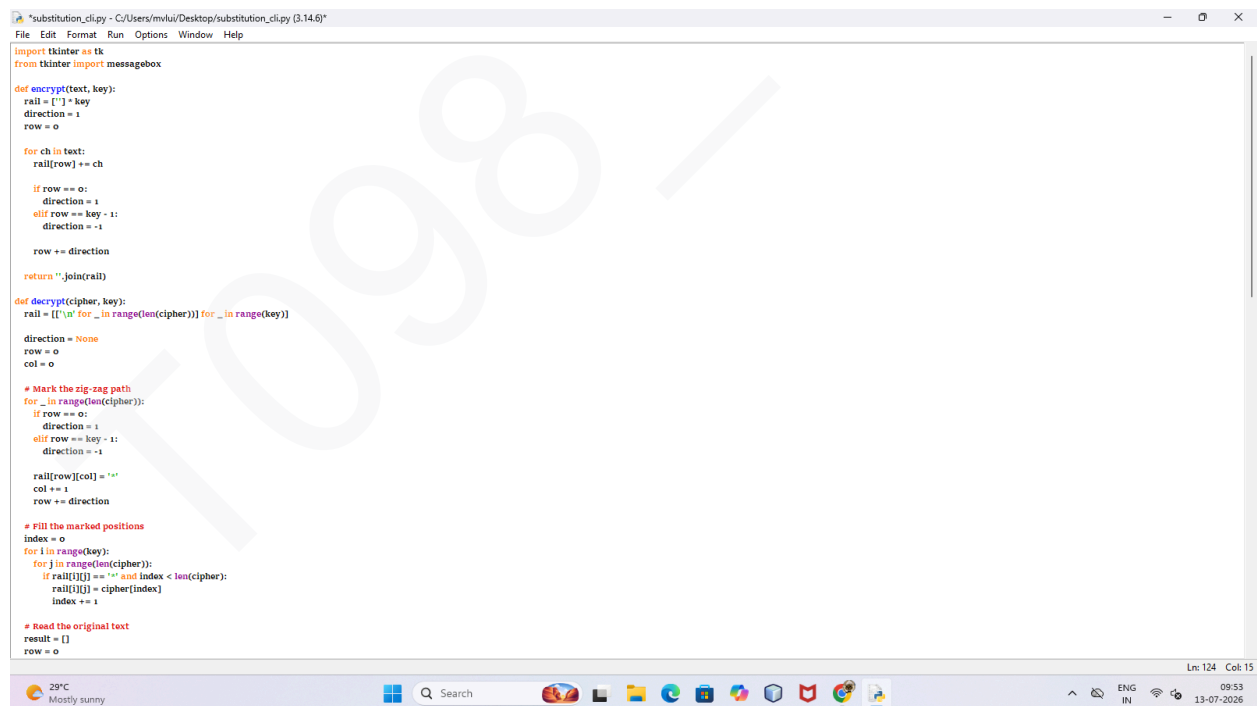


```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/mvlui/Desktop/substitution_cli.py =====
Enter Message : Vikas Pal
Enter Key : 3
Encrypted : Vslia akP
Decrypted : Vikas Pal

rest
row
col
for:
if
if
re
co
re
retu
text =
key =
cipher
print(
plain
print(
```

GUI:-



```
import tkinter as tk
from tkinter import messagebox

def encrypt(text, key):
    rail = ""
    direction = key
    row = 0

    for ch in text:
        rail[row] += ch

        if row == 0:
            direction = 1
        elif row == key - 1:
            direction = -1

        row += direction

    return "".join(rail)

def decrypt(cipher, key):
    rail = [" " for _ in range(len(cipher))]
    direction = None
    row = 0
    col = 0

    # Mark the zig-zag path
    for _ in range(len(cipher)):
        if row == 0:
            direction = 1
        elif row == key - 1:
            direction = -1

        rail[row][col] = "+"
        col += 1
        row += direction

    # Fill the marked positions
    index = 0
    for i in range(key):
        for j in range(len(cipher)):
            if rail[j][i] == "+" and index < len(cipher):
                rail[j][i] = cipher[index]
                index += 1

    # Read the original text
    result = []
    row = 0
```

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```
"substitution_cli.py - C:/Users/mvlu/Desktop/substitution_cli.py (3.14.6)"
File Edit Format Run Options Window Help

# Read the original text
result = []
row = 0
col = 0

for _ in range(len(cipher)):
    if row == 0:
        direction = 1
    elif row == key - 1:
        direction = -1

    result.append(result[row][col])
    col += 1
    row += direction

return "".join(result)

def enc():
    try:
        text = msg.get()
        key = int(key_entry.get())

        if key < 2:
            messagebox.showerror("Error", "Key must be greater than 1.")
            return

        cipher = encrypt(text, key)

        out.config(text="Encrypted:" + cipher)

        msg.delete(0, tk.END)
        msg.insert(0, cipher)

    except ValueError:
        messagebox.showerror("Error", "Please enter a valid key.")

def dec():
    try:
        text = msg.get()
        key = int(key_entry.get())

        if key < 2:
            messagebox.showerror("Error", "Key must be greater than 1.")
            return

        plain = decrypt(text, key)

        out.config(text="Decrypted:" + plain)

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```

```
"substitution_cli.py - C:/Users/mvlu/Desktop/substitution_cli.py (3.14.6)"
File Edit Format Run Options Window Help

msg.delete(0, tk.END)
msg.insert(0, cipher)

except ValueError:
    messagebox.showerror("Error", "Please enter a valid key.")

def dec():
    try:
        text = msg.get()
        key = int(key_entry.get())

        if key < 2:
            messagebox.showerror("Error", "Key must be greater than 1.")
            return

        plain = decrypt(text, key)

        out.config(text="Decrypted:" + plain)

        msg.delete(0, tk.END)
        msg.insert(0, plain)

    except ValueError:
        messagebox.showerror("Error", "Please enter a valid key.")

root = tk.Tk()
root.title("Mail Fence Cipher")
root.geometry("400x230")
root.resizable(False, False)

tk.Label(root, text="Enter Message:", font=("Arial", 11)).pack(pady=5)
msg = tk.Entry(root, width=35, font=("Arial", 11))
msg.pack()

tk.Label(root, text="Enter Key:", font=("Arial", 11)).pack(pady=5)
key_entry = tk.Entry(root, width=10, font=("Arial", 11))
key_entry.pack()

tk.Button(root, text="Encrypt", width=12, command=enc).pack(pady=5)
tk.Button(root, text="Decrypt", width=12, command=dec).pack(pady=5)

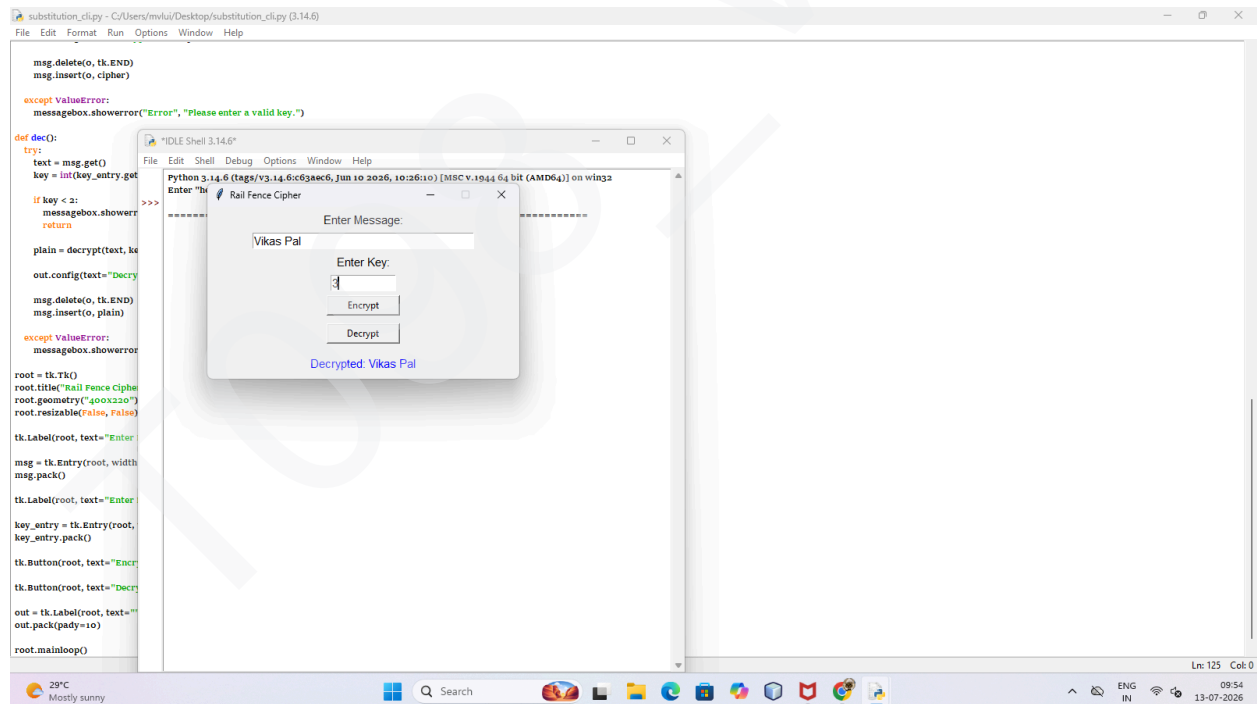
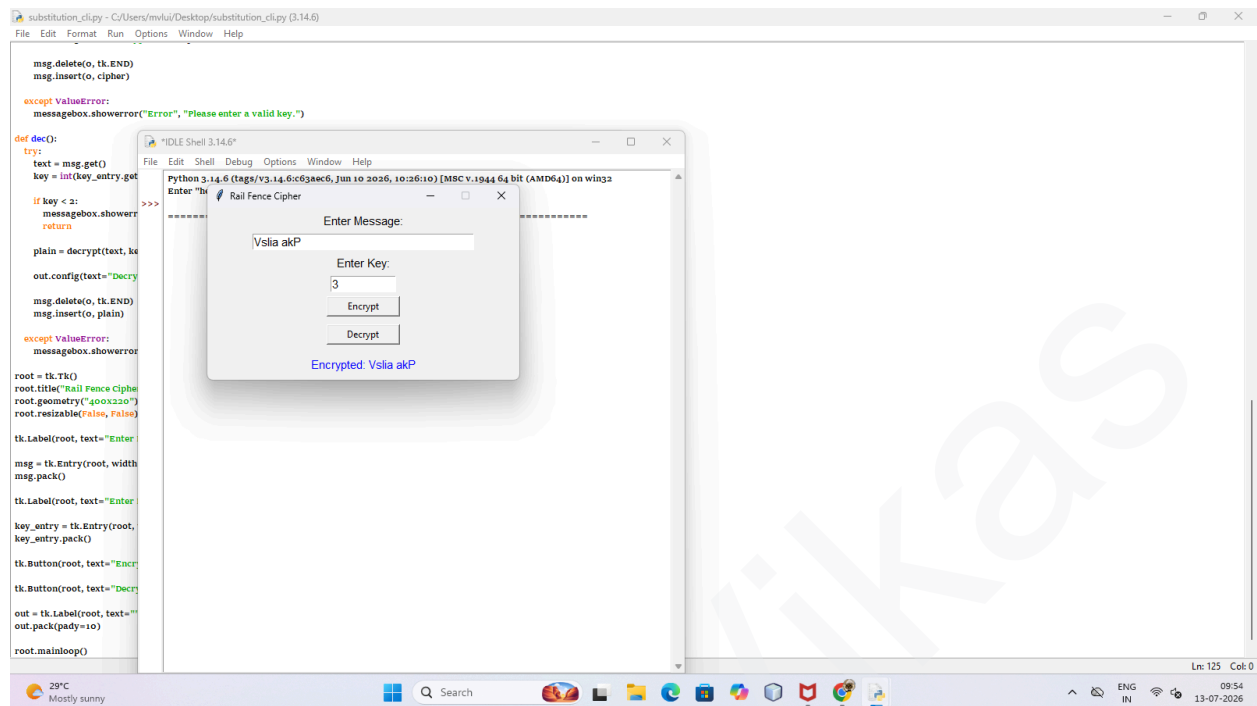
out = tk.Label(root, text="", font=("Arial", 11), fg="blue")
out.pack(pady=10)

root.mainloop()
```

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Output:-



Vikas Pal
T098